ATLAS HR ANALYTICS DASHBOARD

Comprehensive Workforce Intelligence Report

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Tool: Microsoft Power BI

Dataset: Atlas Company HR Data

1. Introduction

HR departments increasingly depend on structured data analysis to make informed decisions about workforce planning, employee well-being, and retention. Without this visibility, patterns that drive costly turnover or declining engagement often go unnoticed until they have already taken hold.

This report documents an end-to-end HR analytics solution built for the Atlas company dataset using Microsoft Power BI. The work covers every stage of the pipeline from raw CSV ingestion and data cleaning, through relational modelling and DAX measure development, to a suite of interactive dashboards. The goal throughout was to surface insights that are genuinely actionable for an HR team, not merely descriptive.

2. Objectives

This project was scoped around four concrete questions that HR leadership typically struggles to answer from raw headcount data alone:

- How are employees distributed across departments, roles, and tenures — and where are the gaps?

- What do performance and satisfaction scores look like over time, and which employees show early warning signs?
- Where is attrition concentrated, and what factors most reliably predict it?
- What data-backed steps can HR take right now to improve retention and workforce composition?

Each dashboard in the final solution maps directly to one of these questions, so findings translate into decisions rather than sitting in a report.

3. Tools & Technologies

The solution was built entirely within Microsoft's analytics stack, keeping the final deliverables portable and maintainable by any analyst already familiar with Power BI.

Microsoft Power BI	Primary environment for data modelling, measure authoring, and dashboard design
Power Query (M)	All data cleaning and transformation logic is applied before data reaches the model
DAX	Calculated measures and KPIs, including the dynamic date table
CSV	Source data format for all employee, performance, and lookup datasets

4. Data Preparation & Transformation

Before any modelling could begin, the raw CSV files required a meaningful amount of remediation. The issues encountered were typical of HR datasets maintained across multiple systems over time — inconsistent casing, orphaned records, and fields that had accumulated a mix of value formats.

4.1 Cleaning

Duplicate records were removed using PerformanceID as the unique key. Blank and null rows were dropped from all lookup tables, and text fields were trimmed of trailing whitespace. Categorical columns with mixed casing "BACHELORS" appearing alongside "Bachelors", for

example, were standardised to a single form so they could be used reliably in filters and relationships.

4.2 Transformation

With the data clean, Power Query was used to promote headers, enforce consistent data types column by column, and remove fields that carried no analytical value. Conditional transformations were applied where business rules required mapping raw values to cleaner labels. The result was a set of well-structured tables ready to slot into a star schema without further adjustment.

5. Data Modelling

The model follows a star schema, with a single fact table at the centre and five dimension tables radiating outward. This structure was chosen because it keeps queries fast, makes relationships easy to reason about, and avoids the fan-trap and chasm-trap issues that crop up in more complex schemas.

Fact Table	PerformanceRating is a central table containing all employee performance metrics
Employee	Core employee attributes: department, role, tenure, travel frequency, attrition status
EducationLevel	Lookup table for education category labels
RatingLevel	Performance rating classifications
SatisfiedLevel	Satisfaction category labels tied to survey response scores
Dim_Date	Custom date dimension generated via DAX (see Section 6)

Relationships between the fact table and each dimension were set up as single-direction filters, which keep cross-report interaction predictable and prevent ambiguous filter propagation.

6. Date Dimension

Rather than importing a pre-built date table, Dim_Date was generated dynamically in DAX using the CALENDAR() function, seeded from the minimum and maximum hire dates present in the

Employee table. This means the date range automatically adjusts if historical records are added or the dataset is refreshed with more recent hires.

The table includes year, month, quarter, week number, and day-level attributes, as well as month and quarter start and end boundaries. These fields support both the time-series charts in the dashboards and any ad hoc date filtering an analyst might apply during exploration.

7. Measures & KPI Design

All DAX measures were centralised in a dedicated Metrics table rather than scattered across the model. This keeps the field list clean and makes it obvious where to look when a measure needs updating.

7.1 Headline KPIs

1,470	1,233	237	16.12%
Total Employees	Active Employees	Inactive Employees	Attrition Rate

7.2 DAX Measures

Measure	DAX Formula
Total_employee	`= COUNT(Employee[EmployeeID])`
Active_employee	`= CALCULATE(COUNT(Employee[EmployeeID]), Employee[Attrition] = "No")`
Inactive_employee	`= CALCULATE(COUNT(Employee[EmployeeID]), Employee[Attrition] = "Yes")`
Attrition_rate	`= DIVIDE([Inactive_employee], [Total_employee])`

CALCULATE handles conditional aggregation cleanly, and DIVIDE is used in place of simple division throughout to avoid divide-by-zero errors surfacing in visuals when filters reduce the denominator to zero.

8. Dashboard Analysis

8.1 Overview Dashboard

The overview provides a one-page summary of the workforce at a glance. The most immediate reading from this dashboard is that an attrition rate of around 16% sits at the higher end of what would be considered acceptable for most industries, and the hiring trend line shows recruitment has been uneven over the reporting period rather than following a consistent growth curve. The Technology and Sales departments account for the largest share of headcount, which shapes where retention efforts have the most scale impact.

8.2 Demographic Analysis

The demographic dashboard breaks down the workforce by age band, marital status, gender, and ethnicity. The majority of employees fall between 18 and 35, and a large proportion are married. Some demographic groups appear underrepresented relative to the workforce as a whole, and salary data layered over these segments shows variation that warrants further investigation — it is not possible from this dataset alone to determine whether those differences reflect legitimate seniority factors or underlying equity issues.

8.3 Performance Tracker

Unlike the other dashboards, the Performance Tracker is designed for individual-level analysis. A slicer allows HR business partners to pull up a single employee and see how their job satisfaction, environment satisfaction, and manager rating scores have trended over time. The practical value here is in spotting decline before it becomes a resignation: an employee whose satisfaction scores have dropped across three consecutive review periods is a different conversation from one who has had a single low score.

8.4 Attrition Analysis Dashboard

This is where the most operationally useful findings sit. Attrition is significantly higher among employees in their first two years, which points to an onboarding or early-engagement problem rather than a general retention problem. Employees who travel frequently for work show elevated attrition rates compared to those in non-travel or rare-travel roles. Certain departments

and job roles account for a disproportionate share of leavers, and the year-on-year trend data makes it possible to assess whether the situation has been improving or worsening over time.

9. Business Recommendations

The findings above point to several specific areas where HR intervention is likely to have the highest return:

- Onboarding quality is the single highest-leverage area. Given that attrition is concentrated in employees with short tenures, a structured 90-day onboarding programme with clear milestones and regular check-ins would directly address the most acute part of the problem.
- Travel policy deserves a second look. The correlation between frequent business travel and higher attrition suggests that the burden of travel may not be distributed equitably or supported adequately. Reviewing which roles genuinely require frequent travel and whether remote alternatives are feasible is a low-cost first step.
- Targeted retention in high-attrition departments. Rather than applying blanket retention incentives, the attrition dashboard makes it possible to focus resources on the specific roles and teams where turnover is actually occurring.
- Early satisfaction monitoring. The Performance Tracker makes it practical to flag employees showing a declining satisfaction trend before they decide to leave. Building a monthly review of declining-score employees into the HR calendar would make this proactive rather than reactive.
- Diversity and pay equity review. The demographic data shows representation gaps and salary variance across groups. A formal pay equity audit, even a lightweight one, would clarify whether the observed differences are explained by seniority and role or require correction.
- Stabilising hiring cadence. The uneven hiring trend visible in the Overview dashboard suggests recruitment planning is reactive. Forecasting headcount needs a quarter ahead, even at the department level would help smooth this out.

10. Conclusion

This project produced a working HR analytics solution that takes Atlas from raw CSV files to a set of interactive, decision-ready dashboards. The data work cleaning, modelling, and measure

development was kept as transparent as possible so that the outputs can be trusted and the model can be extended if the underlying data changes.

The most important finding is not a single number but a pattern: attrition at Atlas is not uniformly distributed. It is concentrated among newer employees, frequent travellers, and specific departments. That means retention spend and HR attention can be directed precisely rather than spread thin across the whole workforce.

The dashboards also create a feedback loop that did not exist before. Satisfaction scores, attrition trends, and hiring data can now be reviewed monthly rather than surfaced only when something has already gone wrong. That shift from reactive to preventive is ultimately what analytics infrastructure is for.